

Cryogenic Resistance vs. Temperature experiment for a Germanium sample using a two-probe configuration.

Aim

To study the variation of resistance of a Germanium (Ge) semiconductor sample with temperature at cryogenic levels using a two-probe configuration, and to determine the bandgap energy (E_g) of the material.

Apparatus

- **Cryostat:** Liquid nitrogen (LN₂) cryostat or Closed-Cycle Refrigerator (CCR) capable of reaching ~77 K.



Figure : Liquid Nitrogen cooled cryostat

- **Sample:** Intrinsic or lightly doped Germanium (Ge) crystal.
- **Vacuum System:** Rotary backing pump and turbomolecular pump (to evacuate the cryostat and prevent ice formation/thermal leaks).
- **Temperature Controller:** e.g., Lakeshore temperature controller with a calibrated sensor (Silicon diode or PT100) and embedded sample heater.

- **Measurement Electronics:** Constant current source and a highly sensitive digital multimeter (or a Source Measure Unit like a Keithley 2400).
- **Connections:** Two-probe wiring (typically phosphor bronze or manganin wire to minimize thermal conductivity).

Theory

Germanium is a semiconductor with a narrow bandgap ($E_g \approx 0.67$ eV at room temperature). The electrical conductivity σ (and consequently the resistance R) of a semiconductor is highly dependent on temperature.

At higher temperatures (the intrinsic region), thermal excitation promotes electrons from the valence band across the bandgap into the conduction band. The carrier concentration n in the intrinsic region varies exponentially with temperature according to the Maxwell-Boltzmann approximation:

$$n \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$

Where:

- E_g = Bandgap energy of Germanium
- k_B = Boltzmann constant (8.617×10^{-5} eV/K)
- T = Absolute temperature in Kelvin

Since resistance R is inversely proportional to carrier concentration ($R \propto 1/n$), the temperature dependence of the resistance is given by:

$$R(T) = R_0 \exp\left(\frac{E_g}{2k_B T}\right)$$

Taking the natural logarithm on both sides yields the Arrhenius equation:

$$\ln(R) = \ln(R_0) + \left(\frac{E_g}{2k_B}\right) \frac{1}{T}$$

By plotting $\ln(R)$ on the y-axis against $1/T$ on the x-axis, the intrinsic region will appear as a straight line. The slope m of this line is used to extract the bandgap:

$$E_g = 2k_B m$$

Note on Two-Probe Configuration:

In a two-probe measurement, the multimeter measures the total resistance of the circuit:

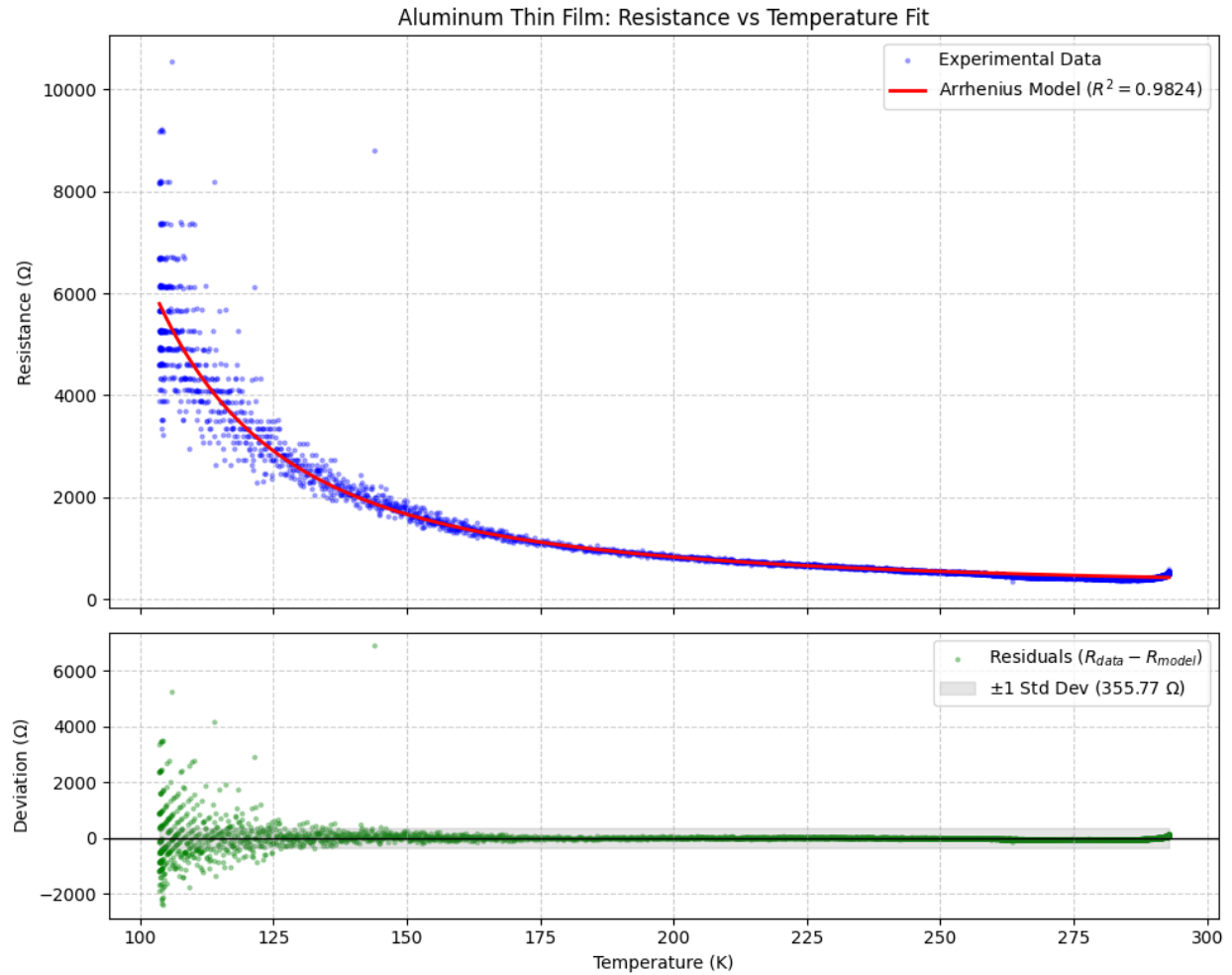
$$R_{\text{measured}} = R_{\text{sample}} + 2R_{\text{contact}} + R_{\text{leads}}$$

While standard precise resistivity measurements utilize a four-probe setup to eliminate contact resistance (R_{contact}), a two-probe setup is sufficient for demonstrating the exponential $R(T)$ trend, provided that at cryogenic temperatures, the sample resistance R_{sample} becomes exponentially large and dominates the parasitic contact and lead resistances.

Procedure

1. **Sample Mounting:** Mount the Ge sample securely on the copper cold finger of the cryostat using a thermally conductive but electrically insulating varnish (e.g., GE Varnish or Apiezon N grease).
2. **Wiring:** Connect the two electrical leads to the Ge sample using silver paste or indium solder. Connect the other ends to the measurement feedthroughs.
3. **Evacuation:** Turn on the rotary and turbo pumps to evacuate the cryostat chamber down to a high vacuum (typically 10^{-4} to 10^{-6} mbar). This is critical to prevent water vapor from condensing on the sample and causing electrical shorting or thermal leaks.
4. **Cooling:** Introduce liquid nitrogen into the cryostat (or turn on the CCR compressor) to cool the sample down to the base temperature (~ 77 K). Allow the system to reach thermal equilibrium.
5. **Measurement Setup:** Configure the SMU to output a small constant current (e.g., $1\mu\text{A}$ to $100\mu\text{A}$). Ensure the current is kept low enough to prevent Joule heating (I^2R) within the highly resistive sample at low temperatures.
6. **Data Acquisition:** Program the temperature controller to slowly warm the sample at a steady sweep rate (e.g., 1 K/min to 2 K/min).
7. **Recording:** Record the Voltage drop (V) and Temperature (T) at regular intervals (e.g., every 2 K). Calculate $R = V/I$. Continue recording until the sample reaches room temperature (~ 300 K).

Observation and Calculations



Data in CSV format is available at : <https://github.com/jajapuramshivasai/Advanced-Experimental-Techniques-in-Quantum-Materials/>

Results

Model Fit : $R^2 = 0.971723$

Estimated bandgap $E_g \approx 2 \cdot \Delta E$: 0.7246 eV

Precautions & Sources of Error

- **Joule Heating:** At cryogenic temperatures (e.g., < 100 K), Germanium becomes highly resistive. Applying too high a measurement current will cause local I^2R heating, raising the sample temperature above the sensor temperature reading.
- **Contact Freezing:** The silver paste or solder contacts may become non-ohmic (forming Schottky barriers) at liquid nitrogen temperatures. This is the primary fundamental limitation of the two-probe setup in this experiment.
- **Thermal Lag:** If the heating/cooling rate is too fast, the temperature sensor (usually mounted on the copper block) and the Ge sample will not be in thermal equilibrium, leading to hysteresis in the R vs. T curve.
- **Light Exposure:** Ge is a photosensitive material. The cryostat optical windows (if any) must be blocked to prevent electron-hole pair generation via the photoelectric effect, which would artificially lower the resistance.